

LABOULAYE, Argentina

WMO: 875340

Lat: 34.1278S

Lon: 63.3675W

Elev: 137

StdP: 99.69

Time Zone: -3.00 (W03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.5	0.1	-8.4	1.9	8.8	-6.5	2.2	7.2	13.4	13.1	11.8	15.3	0.6	230	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	34.6	21.8	33.0	21.4	31.6	21.2	25.0	30.3	24.0	29.3	23.1	28.3	5.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.6	28.1	22.4	17.4	26.9	21.5	16.4	25.9	77.1	30.1	73.0	29.1	69.3	28.3	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.0	9.7	-4.6	38.1	1.8	2.0	-5.9	39.5	-7.0	40.7	-8.0	41.8	-9.4	43.3		
(5)			-5.1	26.8	1.6	1.1	-6.3	27.6	-7.2	28.2	-8.1	28.8	-9.3	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	23.7	22.7	20.4	17.2	13.3	10.0	9.4	11.2	14.1	17.2	20.6	22.8		
	DBStd	6.02	2.95	3.10	3.29	3.77	3.61	2.82	3.40	3.81	3.72	3.38	3.39	3.11		
	HDD10.0	136	0	0	0	2	11	34	52	29	7	1	0	0		
	HDD18.3	1216	2	4	17	65	162	251	277	223	133	62	17	4		
	CDD10.0	2633	426	354	324	216	111	33	67	131	223	318	396			
	CDD18.3	673	169	124	82	29	5	0	1	3	8	26	85	141		
	CDH23.3	6498	1663	1061	660	254	28	1	4	47	134	322	879	1444		
	CDH26.7	2536	730	408	217	56	2	0	1	14	32	97	343	636		
(14) Wind	WSAvg	2.9	3.1	2.4	2.5	2.4	2.2	2.1	2.4	3.2	3.7	4.0	3.8	3.4		
(15) Precipitation	PrecAvg	855	120	104	133	80	32	17	16	19	43	90	108	122		
	PrecMax	1387	331	356	416	260	174	91	110	106	168	232	246	377		
	PrecMin	542	5	17	14	1	0	0	0	0	7	25	9			
	PrecStd	192	65	71	79	55	34	24	18	25	36	58	56	73		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	35.1	33.3	30.6	26.3	21.7	22.2	28.5	30.1	32.6	35.0	36.7		
		MCWB	21.5	23.3	22.3	19.8	18.5	14.5	15.6	17.4	17.4	19.4	20.4	21.5		
	2%	DB	34.5	32.7	30.9	28.2	23.5	19.2	19.3	23.5	26.9	29.5	32.6	34.2		
		MCWB	22.5	23.1	22.3	19.2	17.1	13.6	12.9	15.0	16.3	18.6	19.4	21.0		
	5%	DB	32.6	31.1	29.1	26.4	21.2	17.5	17.4	20.5	24.4	26.7	30.5	32.3		
		MCWB	22.1	22.5	21.0	18.5	15.8	12.4	11.9	12.7	14.8	17.1	19.0	20.6		
	10%	DB	30.9	29.1	27.2	24.3	19.1	15.8	15.6	18.2	22.2	24.5	28.3	30.2		
		MCWB	21.8	21.7	20.2	17.9	15.0	11.2	11.3	11.7	13.8	16.0	18.1	20.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	26.2	25.2	23.2	20.2	17.7	17.0	19.0	20.0	22.0	22.7	25.4		
		MCDB	31.5	31.7	30.3	26.6	23.5	19.6	20.5	23.6	26.7	29.8	29.4	31.7		
	2%	WB	24.9	24.8	23.6	21.4	18.6	15.3	15.2	16.7	17.7	20.1	21.3	23.5		
		MCDB	30.4	29.6	28.0	25.4	21.7	17.2	17.2	20.9	23.7	26.2	28.3	29.6		
	5%	WB	23.9	23.7	22.4	20.0	17.1	13.5	13.7	14.9	16.1	18.8	20.3	22.4		
		MCDB	30.0	28.4	26.9	23.5	20.1	16.1	16.0	18.3	21.4	24.1	27.3	28.7		
	10%	WB	22.8	22.9	21.3	18.8	15.7	12.0	12.1	12.9	14.9	17.5	19.4	21.4		
		MCDB	28.4	27.6	25.2	23.0	18.3	14.9	14.6	17.5	20.4	22.7	25.9	27.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	11.2	11.4	11.5	10.7	11.5	11.3	12.7	13.2	12.0	12.9	12.8		
		MCDBR	15.2	13.8	14.1	15.0	13.4	14.0	14.1	16.3	17.4	16.3	16.7	16.2		
		MCWBR	5.6	5.5	6.1	7.1	7.3	8.3	8.5	8.6	8.5	7.5	6.3	5.7		
	5% WB	MCDBR	11.9	11.3	11.6	11.5	9.6	11.1	10.5	13.1	14.0	13.1	13.5	12.6		
		MCWBR	5.9	5.3	5.8	6.3	5.7	7.6	7.4	7.8	7.7	7.1	5.9	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.403	0.390	0.377	0.365	0.347	0.324	0.335	0.375	0.423	0.403	0.393	0.395		
	taud		2.342	2.382	2.417	2.422	2.422	2.484	2.422	2.281	2.147	2.267	2.326	2.343		
	Ebn at Noon		935	925	899	851	810	809	814	823	836	902	938	946		
	Edh at Noon		134	124	111	99	87	77	85	111	143	137	135	134		
(44) All-Sky Solar Radiation	RadAvg		7.37	6.35	5.26	3.80	2.68	2.44	2.59	3.52	4.68	5.68	6.93	7.54		
	RadStd		0.46	0.50	0.43	0.46	0.30	0.29	0.25	0.34	0.40	0.52	0.41	0.51		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.50	+0.73	N/A	N/A	N/A

Nomenclature: See separate page